

Osama Abo-Bakr

AI Engineer & Solution Architect

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PROFESSIONAL SUMMARY

AI Engineer & Solution Architect with 2+ years of hands-on experience designing and deploying production-grade AI systems. Specialized in LLM applications, RAG pipelines, multi-agent frameworks, on-device detection models, and workflow automation (n8n). Proven track record building end-to-end AI platforms from MVP to full production — including multi-channel customer support systems, enterprise chatbots, eKYC verification systems, and intelligent data analysis tools. Top-Rated Upwork freelancer with 100% Job Success Score and 5-star client ratings across 10+ delivered projects. Graduate of the Faculty of Artificial Intelligence, Menoufia University.

PROFESSIONAL EXPERIENCE

ML Engineer | Thndr

January 2026 – Present | Cairo, Egypt

- Built an alternative eKYC OCR pipeline (YOLO-based field detection + PaddleOCR v5 + rule-based post-processing), improving field accuracy by up to 22x and cutting latency ~10x (3.3–3.8s → 0.2–0.4s) versus the existing production system.
- Fine-tuned PaddleOCR v5 on domain-specific Arabic national ID data, driving P90 character error rate to 0 across all fields (up to 99.6% field-level accuracy) while holding per-field inference at 3–7ms — boosting overall pipeline throughput.
- Resolved a multi-line field extraction failure by training a dedicated YOLO detection model to separate multi-line text regions, cutting character error rate from 58% to 11%.
- Built a passport eKYC pipeline from scratch (detection → OCR → MRZ extraction → parsing), reaching 1.77% average character error rate on a 2,000-image test set.
- Designed an AI Fundamental Analysis agent for automated financial insight generation and deployed an on-device detection model for low-latency mobile inference.

AI Engineer & Solution Architect | NexAI

2024 – Present | Riyadh, Saudi Arabia (Remote)

- Architected the core AI platform from MVP to production, delivering multi-channel customer support (WhatsApp, Telegram, Instagram, Messenger) serving 500+ daily conversations.
- Designed scalable microservices architecture for chat, knowledge base, analytics, and ticketing — reducing system response time by 40% through async processing and caching.
- Built AI Agent pipelines and RAG systems using LangChain and Pinecone, cutting manual support workload by 60% via intelligent automated responses.
- Integrated voice-enabled chat (ElevenLabs TTS), expanding platform capabilities across 3+ languages.
- Implemented role-based authentication, analytics dashboards, and knowledge-base tools — adopted by 5+ business teams across the organization.
- Collaborated with product managers and business stakeholders to translate requirements into scalable AI solutions, driving product direction from technical architecture to deployment.

Freelance AI Developer | ML Engineer | Upwork

2023 – Present | Remote (Global Clients)

- Delivered 10+ AI solutions for SaaS and e-commerce platforms — earning Top-Rated status, 100% Job Success Score, and 5-star feedback, generating \$15,000+ in revenue.
- Built AI-powered chatbots with recommendation systems and multilingual support, improving client user engagement by 35% for Arabic e-commerce platforms.
- Developed AI automation workflows using LangChain and n8n, reducing clients' manual operational overhead by 15+ hours/week.
- Created an advanced image search engine using computer vision (OpenCV, deep learning) for precise visual content retrieval across large product catalogs.

Independent Contractor – AI Developer | Nexus Mundus Equity LLC

Dec 2024 – Feb 2025 | Remote

- Developed AI chatbot features (order tracking, recommendations, FAQ, cart reminders) for Arabic e-commerce platform, completing all milestones within the 1.5-month timeline.

- Managed full API integration and chatbot deployment lifecycle, ensuring IP compliance and smooth handoff to the client team.

EDUCATION

B.Sc. in Artificial Intelligence (Machine Intelligence Major) | Faculty of Artificial Intelligence, Menoufia University

2022 – 2026 | Cairo, Egypt | Expected Graduation: 2026

- HackerRank: Top 98th Python Developer in Egypt | #1 Python & Data Structures at Menoufia University
- Kaggle Expert | Active in ML competitions and published datasets
- Led faculty projects: Face Recognition Attendance System & AI Essay-Correction System

TECHNICAL SKILLS

AI & LLM: LangChain · LangGraph · LlamaIndex · CrewAI · OpenAI GPT-4 · Hugging Face · Fine-Tuning (LoRA/QLoRA)

ML & Deep Learning: PyTorch · TensorFlow · Scikit-learn · OpenCV · NLP · Transformers

RAG & Vector DBs: Pinecone · Weaviate · FAISS · ChromaDB · Milvus · pgvector

Automation: n8n · Make · Custom API Integrations · Workflow Design

Backend & APIs: FastAPI · REST APIs · Microservices · Async Processing

Infrastructure: Docker · AWS · GCP · Railway · Vercel · CI/CD

Data & Databases: PostgreSQL · Redis · Pandas · NumPy · Streamlit

Frontend: Next.js · React · TypeScript · Tailwind CSS

KEY PROJECTS

eKYC OCR & Field-Detection Pipeline — Thndr

Rebuilt the eKYC OCR pipeline with YOLO field detection and fine-tuned PaddleOCR v5 (Arabic), cutting latency ~10x and boosting field accuracy up to 22x vs. the prior system; also shipped a passport pipeline from scratch.

Tech: YOLO · PaddleOCR v5 · Python · OCR · Computer Vision · Face Matching · Fraud Detection

Multi-Channel AI Customer Support Platform

Production-grade AI support system handling WhatsApp, Telegram, Instagram, and Messenger. Integrated RAG pipelines, voice chat (ElevenLabs), ticketing, and analytics dashboards.

Tech: FastAPI · LangChain · PostgreSQL · Redis · Pinecone · Docker · ElevenLabs

SaaS RAG Chatbot Application

Enterprise document Q&A platform — ingests up to 30 documents, performs hybrid search with metadata extraction, supports multimodal inputs, and maintains conversation memory.

Tech: Python · LangChain · Weaviate · FastAPI · Google API · React · Node.js

Real-Time Stock Market Analysis Agent

Multi-agent system using CrewAI: News Fetcher → Sentiment Analyzer → Technical Indicator Engine → Trading Signal Generator. Processes live financial data and outputs actionable predictions.

Tech: CrewAI · GPT-4 · Finnhub API · yFinance · Pandas · LangChain

InstaMasr – Government Services Platform

AI-powered platform streamlining Egyptian government service access. Features intelligent chat assistant, automated form processing, Arabic RTL support, and secure JWT authentication.

Tech: Next.js 14 · FastAPI · LangChain · TypeScript · Tailwind CSS · JWT

ADDITIONAL EXPERIENCE & ACHIEVEMENTS

Python Lead & Instructor | AlgoMaster (NPO)

2022 – 2023

- Led the Python programming department — prepared curriculum, mentored 20+ students, and organized competitive programming contests.

Research Assistant | Menoufia University

2023 – 2024

- Built a deep learning model for breast cancer detection from X-Ray images — classifying benign vs. malignant and localizing cancerous cells.

LANGUAGES

Arabic (Native) | English (Conversational)